6. Quanta of Space

The last chapter closed with the solutions of the Wheeler– DeWitt equation discovered by Jacobson and Smolin. These solutions depend on lines that close on themselves, or loop. What does it all mean?

Remember Faraday's lines – the lines which carry the electric force and which, in Faraday's vision, fill space? The lines from which the concept of 'field' originates? Well, the closed lines that appear in the solutions of the Wheeler–DeWitt equation are Faraday lines of the gravitational field.

But two new ingredients are now added to Faraday's ideas.

The first is that we are dealing with quantum theory. In quantum theory, everything is discrete. This implies that the infinitely fine, continuous spiderweb of Faraday's lines now becomes similar to a real spiderweb: it has a finite number of distinct threads. Every single line determining a solution of the Wheeler–DeWitt equation describes one of the threads of this web.

The second new aspect, the crucial one, is that we are speaking of gravity and, therefore, as Einstein understood, we are not speaking of fields immersed in space but of the very structure of space itself. Faraday's lines of the quantum gravitational field are the threads of which space is woven.

At first, the research was focused on these lines and how they could 'weave' our three-dimensional physical space. Figure 6.1 represents an early attempt to give an intuitive idea of the discrete structure of space which would result from this.

Soon after, thanks to the intuitions and mathematical ability of young scientists such as the Argentine Jorge Pullin and the Pole Jurek Lewandowski, it became clear that the key to understanding the physics of these solutions lies in the points where these lines intersect. These points are called nodes, and the lines between nodes are called links. A set of intersecting lines forms what is called a graph, that is to say, a combination of nodes connected by links, as in figure 6.3.

A calculation, in fact, demonstrates that, without nodes, physical space has no volume. In other words, it is in the nodes of the graph, not in the lines, that the volume of space 'resides'. The lines 'link together' individual volumes sitting at the nodes.

Getting to a full clarification of the resulting picture of quantum spacetime took years. It was necessary to transform the ill-defined mathematics of the